



Data Visualization Lab
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Traffic Causal Relationship Analysis with Causal Density

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Motivation

- In traffic congestion analysis, many studies have been presented to provide **new insight for reducing traffic congestion**.
- In many fields, causal analysis has a significant impact on decision making and recent **urban data analysis study** has attempted to discover the **causal relationship**.

Causal Calculation

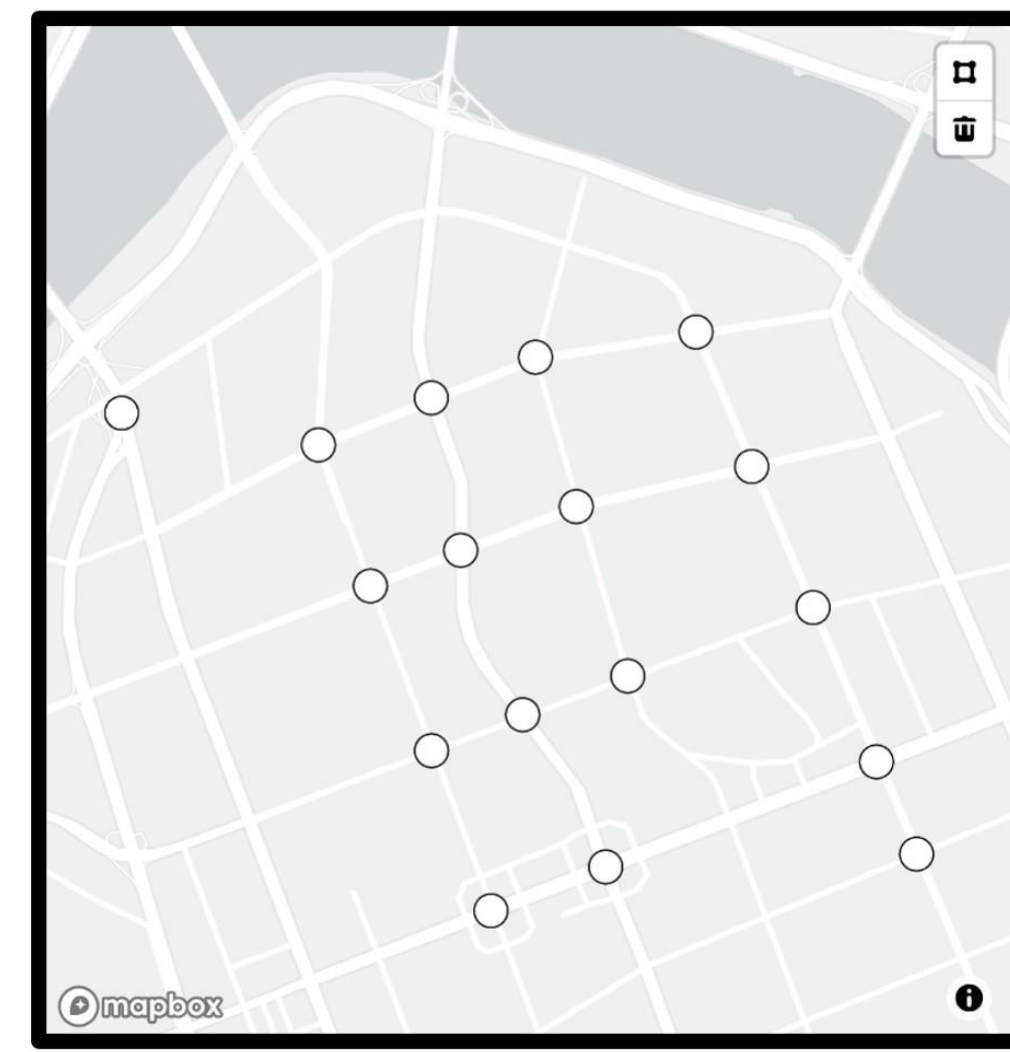
Granger causality test: $F_{\{Y \rightarrow X\}} = \ln \frac{\text{var}(\varepsilon'_x(t))}{\text{var}(\varepsilon_x(t))}$

- Causality implies that one event induces another.
- We use **Granger Causality test** to analyze the causal relationship between two variables for causality analysis.

Causal Density: $CD(X) = \frac{1}{n(n-1)} \sum_{i \neq j} F_{\{X_i \rightarrow X_j\}}$

- Since Granger Causality test does not measure **the strength of causal relationships**, we use **Causal Density**.
- The Causal Density considers the **variance ratio among all elements** in the framework.

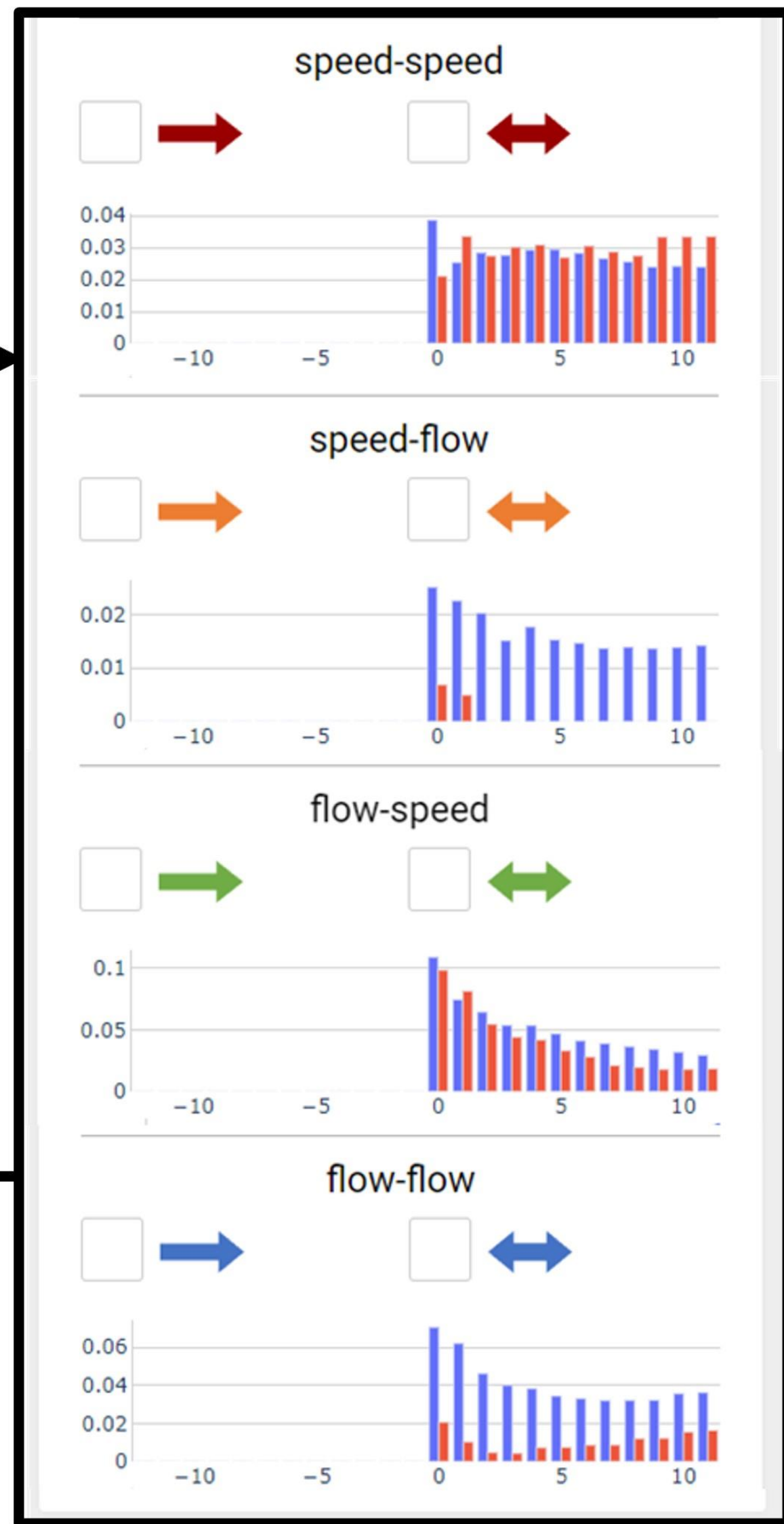
Traffic Network



Traffic Causality Analysis

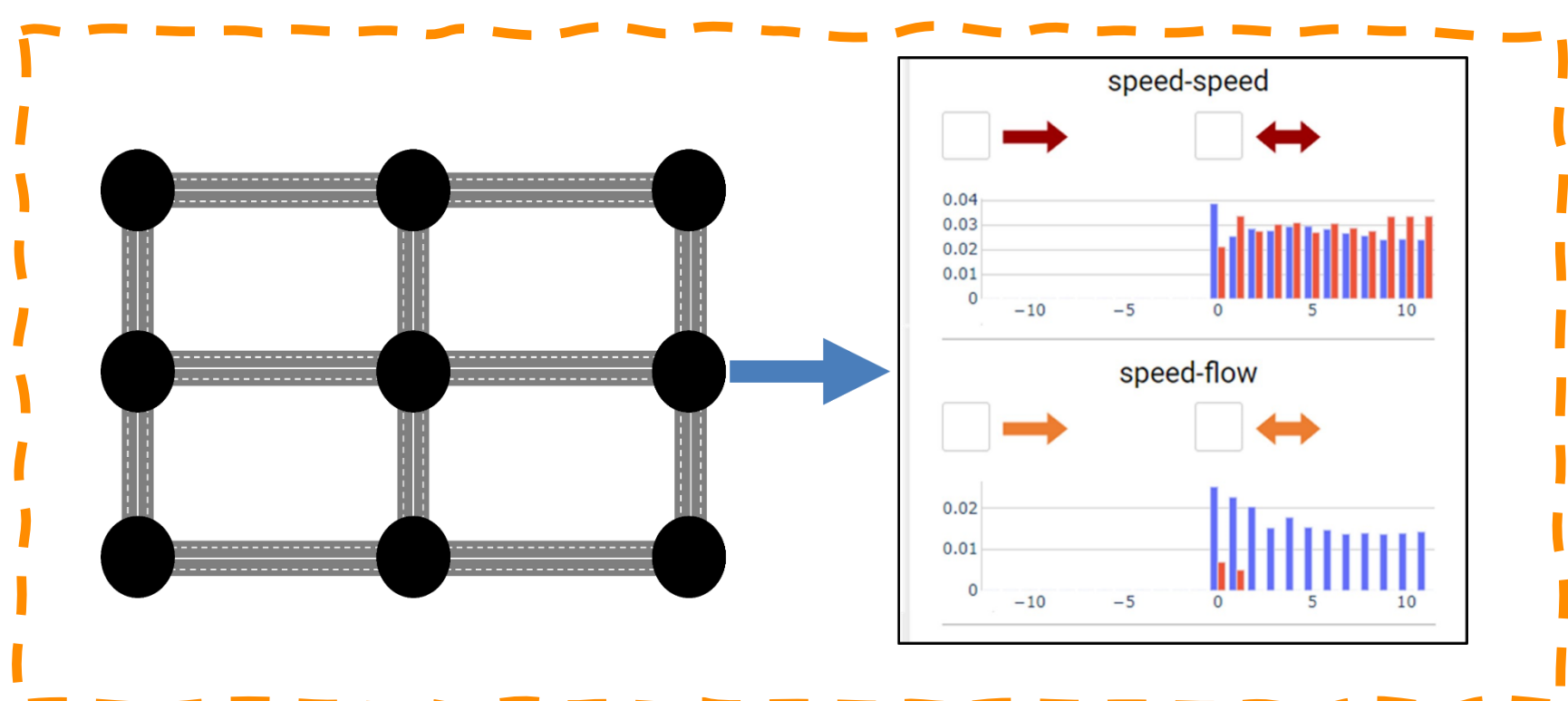


Causal Calculation

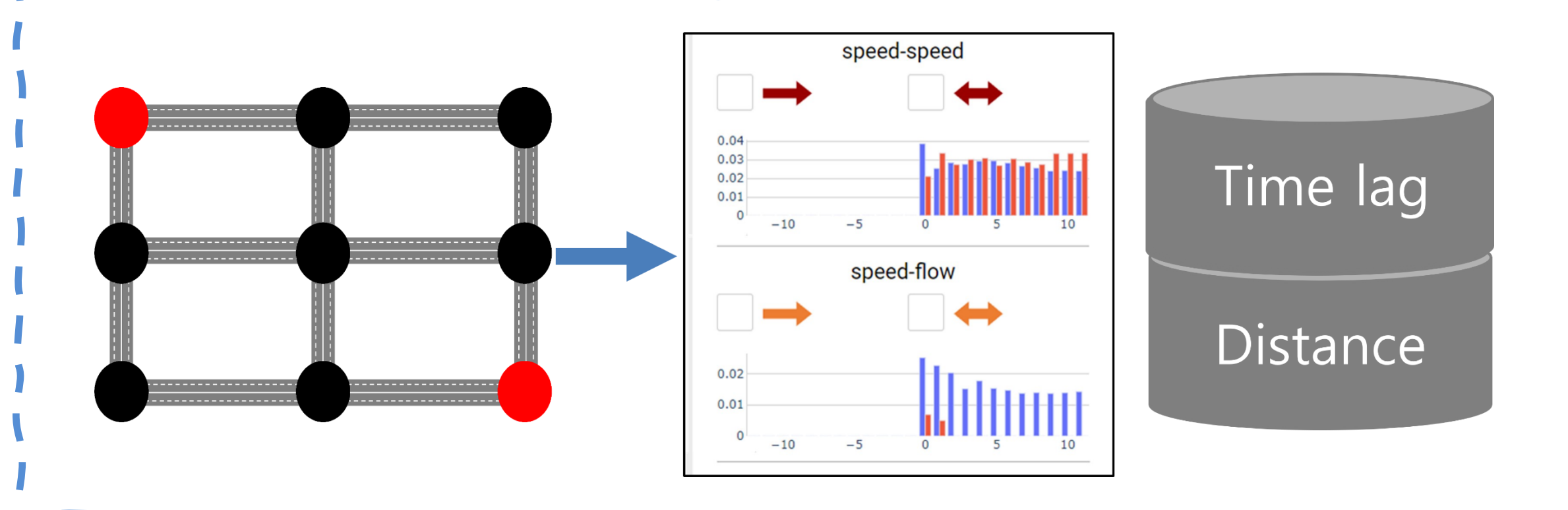


Our Visual Analytics Framework

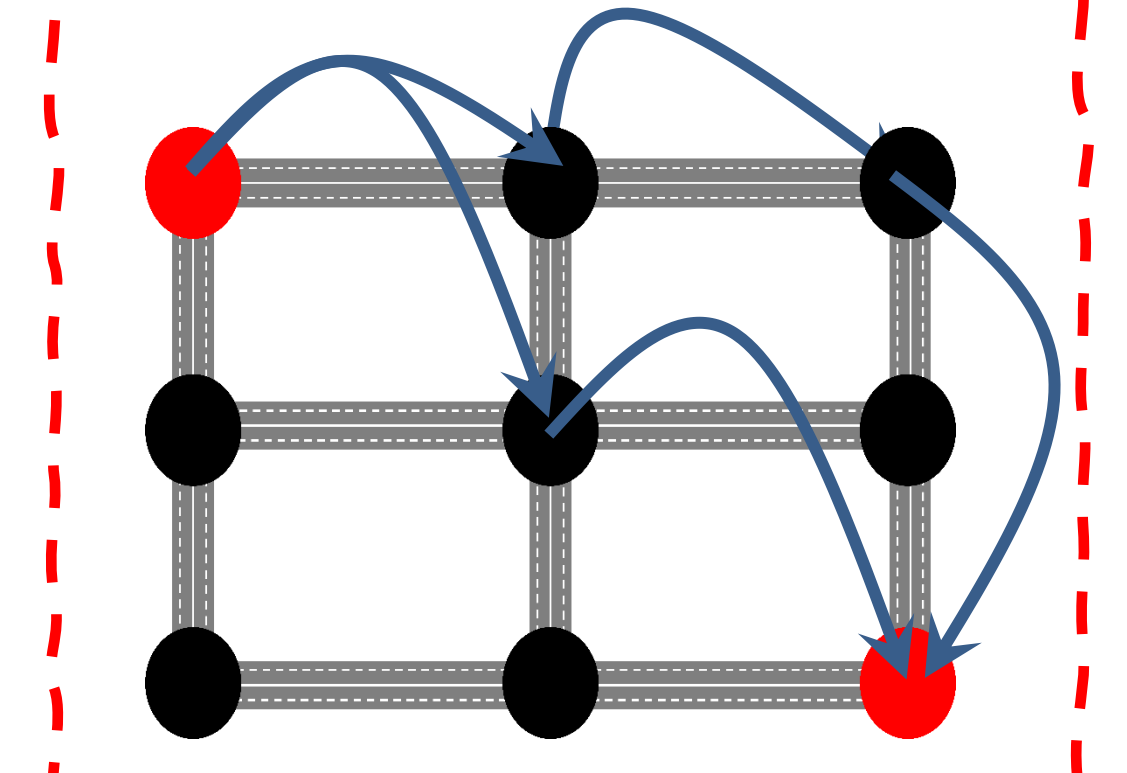
Analysis Process



Finding Significant Causality in Traffic



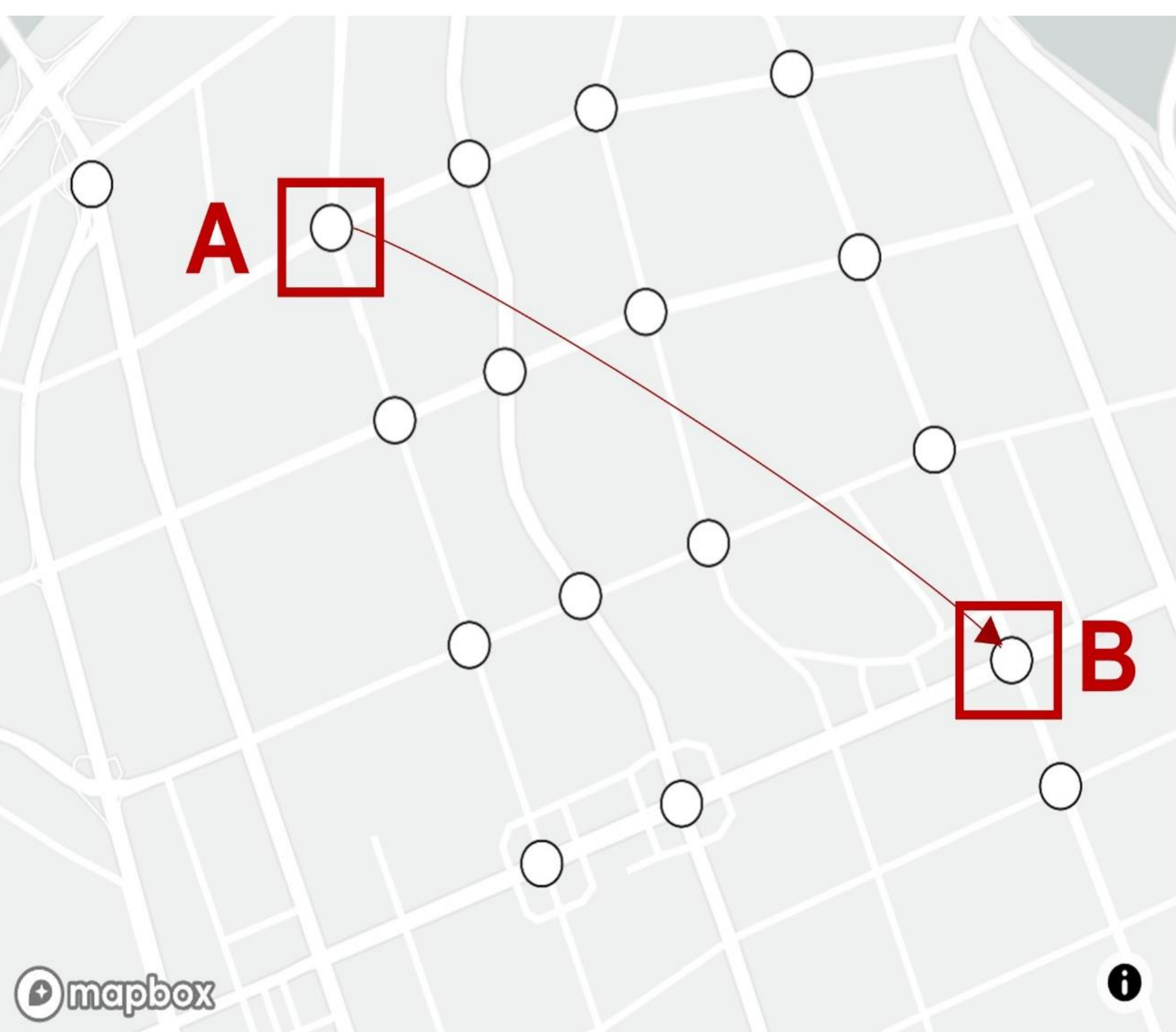
Traffic Casual Analysis with traffic network
by using Time lag and Distance



Visual Analytics Result

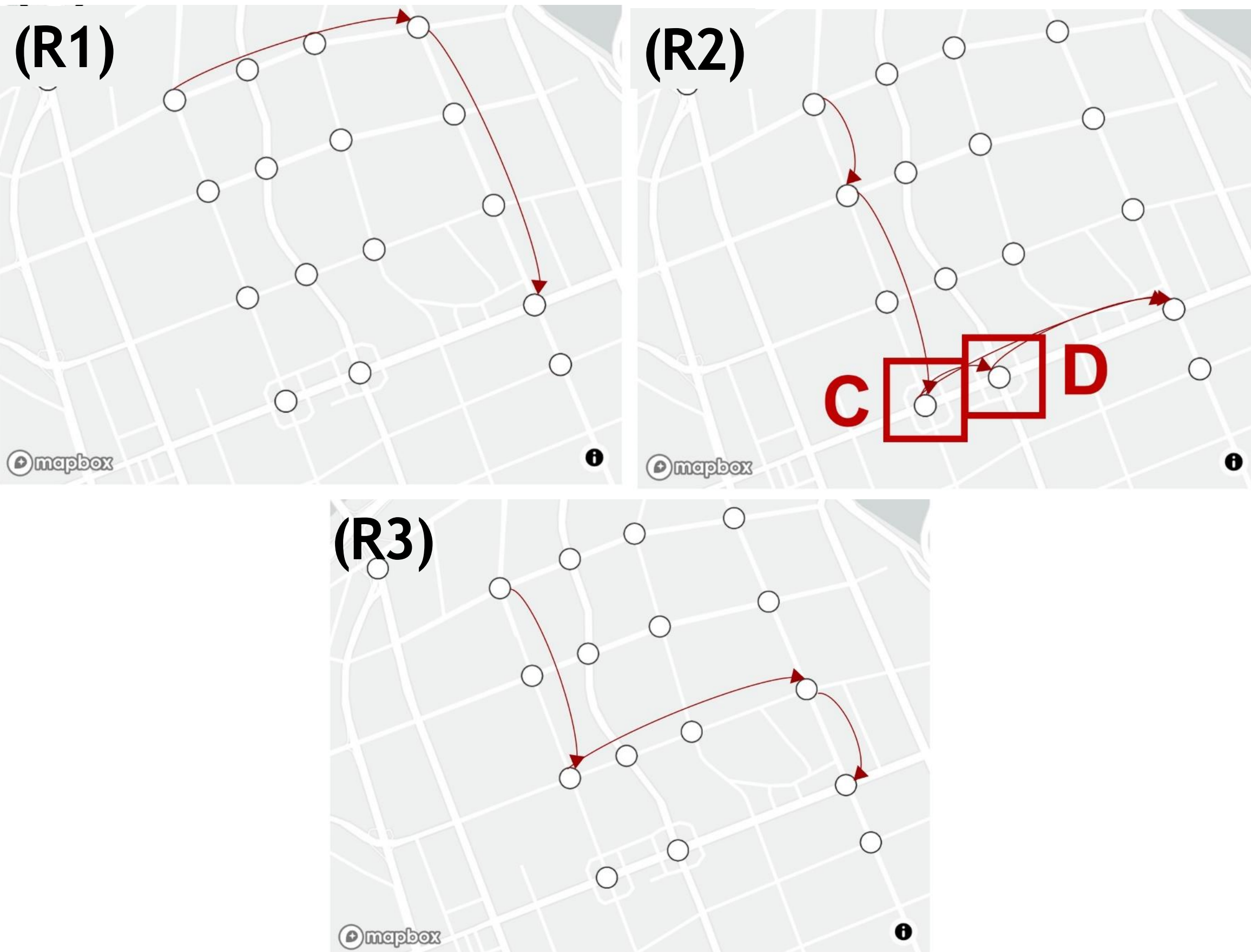
Case Study

(a) High strength causal relationship



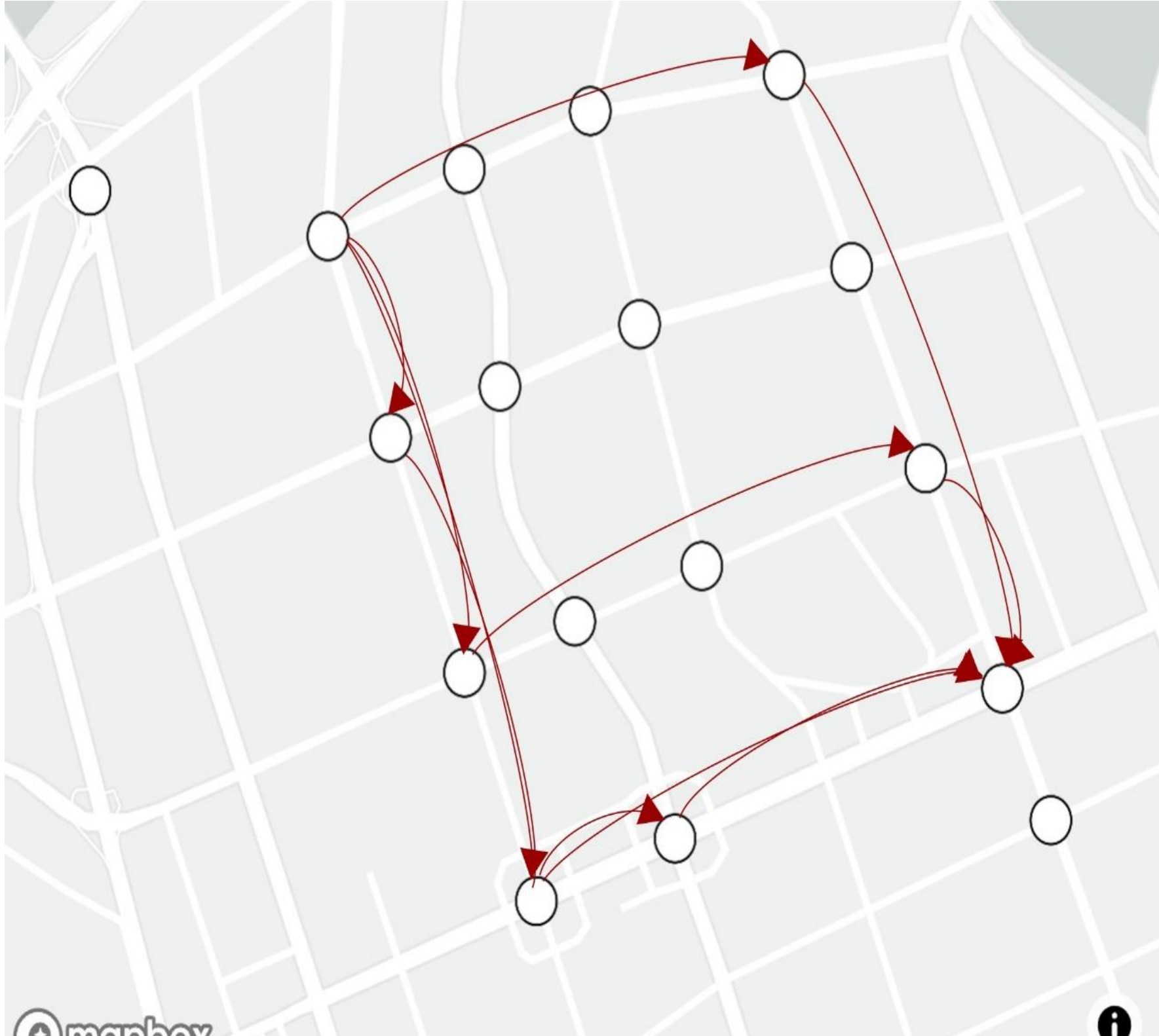
Traffic Causal We found that a **high-strength causal relationship** exists between two **distant nodes**, A and B in Korea road.

(b) Analysis of causal path between node A and B



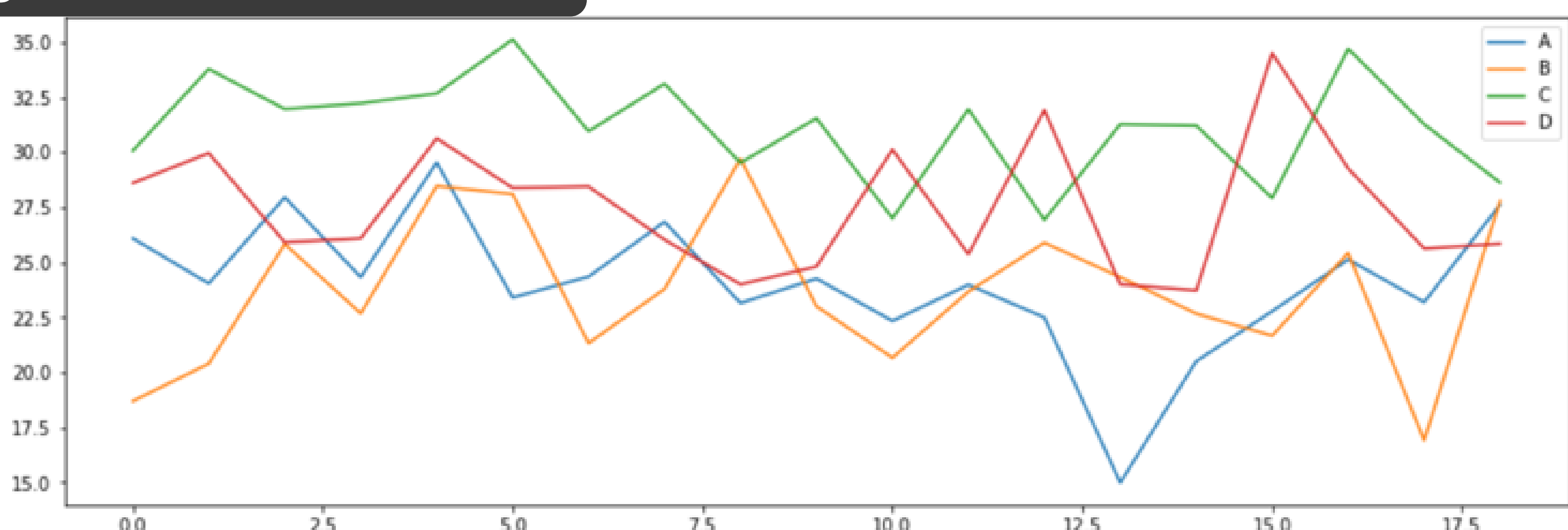
Traffic Causality Analysis We found that a **three high-strength route** between two **distant nodes**, A and B.

(c) Visual Analytics Result



Result (c) as a result of the causal analysis, a **high causal pathway of R1, R2 and R3** was found between nodes A and B, and this **supports the existence of a high causal relationship** directly connected between nodes A and B.

Significant of Result



The original data on vehicle speed in Gangnam traffic

Average speed graph in Figure 3 shows that nodes C, D, and B also **decrease at intervals of 5 minutes** after the speed value starts to decrease at node A.

Conclusion

- In this paper, we introduced a traffic congestion analysis technique based on causal relationships and a novel traffic causal analysis framework supporting traffic congestion tracking and decision-making for traffic policy.
- In addition, a map projection function was included to check how the traffic causal relationship tree affects the traffic network, allowing analysts to discover new perspectives on traffic congestion analysis with causal relationships.